Gear Tube

User manual and parameter reference · 0.1.1

A responsive tube saturator with four tube characters, bias, sag and parallel blending.

Quick start

1. Insert Gear Tube on an audio track and start at low Drive.
2. Choose a tube, add Drive and adjust Output to compare levels.
3. Arm Auto Gain, then move a sound knob to learn a matching output; open HELP for each control.

Editor window

1200 × 785 fixed with LAB closed. LAB expands the editor horizontally. Dimensions are logical editor pixels, before host window chrome. HELP adds 92 pixels of height (Gear adds a one-pixel divider too). Allow enough usable screen space at your display scale. Press HELP again to close the help area.

Operation

Drive adds level and saturation; Output attenuates the wet path before Mix. At Mix 0% you hear the latency-aligned dry signal. Tube selection and switches do not start Auto Gain.

LAB shows the transfer curve, detectors and spectra. It expands the window beside the hardware view; close it to return to the normal width.

Auto Gain is an armed workflow, not continuous normalization. Only direct user changes to sound knobs start learning while it is enabled; it holds for eight seconds after the last such edit and then idles. Output is excluded so you can trim it while idle. Turning Auto Gain on only arms it. Switching it off stops learning. Presets do not store Auto Gain; session recall retains the switch and Output but starts idle. Offline rendering uses the saved output.

Controls

- **Tube**: 12AX7, EL34, KT88, 300B.
- **Drive**: 0 to +40 dB into the stage.
- **Bias**: cold leans the operating point into cutoff (more even harmonics, earlier compression), hot toward grid conduction (bite sooner).
- **Sag**: how much the supply's ceiling comes down under sustained level, of the tube's own depth.
- **Pair**: push-pull.
- **Tone**: the stage's low-pass corner, 3 to 20 kHz.

- **Mix**, **Output**.
- **Link**: the channels' detectors follow the louder channel.
- **Quality**: Standard (2x oversampling with ADAA, about 1 ms of latency), High (4x), Ultra (8x), each longer. Reported to the host.
- **Bypass**.
- **Debug** (top bar): the dynamic curve live. The curve in use over the family's rest curve, the rest curve shifted a knee toward cutoff and the one at full excess; the last input samples scattered on it as (in, out) dots; the operating point (a silent input); the bias, excess and sag detectors over the last seconds; the output's spectrum against the input's; the numbers.
- The tubes glow with the signal (dim while quiet, full when it is loud or the grid conducts: the Bias help says how).

Controls and operation

TUBE

Which tube the signal goes through: 12AX7 (preamp triode, high gain, even harmonics first), EL34 (pentode, harder knee, odd harmonics), KT88 (beam tetrode, big clean headroom, firm low end) or 300B (single-ended triode, soft and even-rich). Click or drag to step; the chosen tube lights up.

12AX7

A preamp triode: high gain, warm and then crunchy. Even harmonics first, a strong bias shift (a small coupling capacitor), no sag, a gentle one-pole tone filter.

EL34

A pentode power tube: a harder knee and more odd harmonics, a medium bias shift, deep sag, and an output transformer's leakage as the tone filter. Use Pair for push-pull.

KT88

A beam tetrode: a late, firm knee and a lot of clean headroom, moderate sag, a heavier transformer core for the low end.

300B

A single-ended directly heated triode: a very soft knee, rich in even harmonics, a weak bias shift, slight sag.

DRIVE

How hard the signal pushes the tube, 0 to +40 dB. Drive 0 passes the signal at its own level and nearly clean; +12 dB brings a full-scale peak to the tube's cutoff knee; past that the tube squashes and the harmonics grow. The wet signal gets the headroom back, so Drive also makes it louder: turn Output down to match.

BIAS

Where the signal sits on the tube's curve, -1 cold to +1 hot. Cold moves toward cutoff: starved, gated, asymmetric and more even harmonics. Hot moves toward grid conduction (the signal pushes the tube's control grid positive and the tube clamps the peaks hard: the crunch), and the tubes glow brighter as it happens. On top of it the stage drifts by itself after loud passages (the coupling capacitor charging) and recovers.

SAG

How far the power supply gives way under sustained loud signal, 0 to 100% of the tube's own depth. The tube's ceiling comes down while it plays hard and recovers in a moment: a slight duck and a softer top, as a tired supply does.

TONE

The corner of the low-pass after the stage, 3 kHz to 20 kHz: first order on the 12AX7 and 300B, second order (a transformer) on the EL34 and KT88. Turn it down to take the fizz off a hard drive.

MIX

Dry against the tube signal, 0 to 100%, after Output: 0% is the input exactly as it came, whatever Drive and Output are. The two are lined up to the sample, so any blend adds up without comb filtering (parallel saturation).

OUTPUT

Takes level off the tube signal, -48 to 0 dB, after the stage and before Mix: it is clean gain (no distortion; Drive, before the stage, is what adds gain and saturation) and the dry signal is never touched by it. Use it to bring the tube signal back to the dry level after Drive, or let Auto Gain set it for you.

AUTO GAIN

Arm Auto Gain, then directly edit a sound knob to match loudness by turning Output. It learns for eight seconds after the last user knob edit, then holds the result while still on. Switches, choices, host automation, presets and session recall do not start learning. Output itself is a manual trim while idle; it is protected during learning. Auto Gain is excluded from presets. Session recall starts idle, and offline rendering uses the saved Output. Working has a pulsing dot and time-left line; idle has a steady dot.

PAIR

Push-pull: a second stage runs on the inverted signal and the two are subtracted. Even harmonics cancel, as in a push-pull output stage, leaving the odd ones: tighter and more symmetrical.

LINK

Stereo link. On, both channels' detectors follow the louder one, so the tube behaves identically on both and the image does not wander. Off, each channel has its own tube.

QUALITY

Standard (2x oversampling), High (4x) or Ultra (8x), all with anti-aliasing of the curve. Higher keeps what the tube adds from folding back as aliasing at high drive, at more CPU and latency. Click to step.

POWER

Switches the stage on or off. Off plays the input as it is (still delayed by the stage's latency, so nothing shifts against other tracks) and the tubes cool down.

POWER LAMP

Lit while the stage is on.

VU METER

The output level. 0 VU is -18 dBFS RMS; the needle follows the signal's average as a VU meter does (99% of a step in 300 ms), full scale at +3 VU. It reads after Mix and Output, and rests when Power is off.

PRESETS

Browse, load and save settings. An asterisk means the sound has changed since the preset was loaded.

HELP

Shows or hides this bar. It says what the control under the mouse is and what it does.

LAB

Opens the lab beside the amplifier: the tube's curve as it moves, the input on it, the detectors, the spectra, and the evidence page that checks each parameter's range and the meter against independent measurements.

BYPASS

The same as the POWER switch: plays the input as it is, in time with the processed signal.

Every parameter

Values and defaults are taken from the plugin. A control's effect can depend on the selected mode; see the operation notes above. Choices keep their saved identity between versions. Knobs and switches can be automated unless marked otherwise.

Parameter	What it does	Values and default
Tube	Tube: 12AX7	12AX7, EL34, KT88, 300B Default: 12AX7 Not host automatable

Parameter	What it does	Values and default
Drive	Drive: +12.0 dB	0.0 dB to +40.0 dB Default: +12.0 dB
Bias	Bias: 0	-1.00 cold to 1.00 hot Default: 0
Tone	Tone: 12.0 kHz	3.00 kHz to 20.0 kHz Default: 12.0 kHz
Sag	Sag: 50%	0% to 100% Default: 50%
Output	Output: 0.0 dB	-48.0 dB to 0.0 dB Default: 0.0 dB
Mix	Mix: 100%	0% to 100% Default: 100%
Pair	Pair: Off	Off to On Default: Off
Link	Link: On	Off to On Default: On
Quality	Quality: Standard	Standard, High, Ultra Default: Standard Not host automatable
Auto Gain	Keeps the tube signal as loud as the dry input by turning the Output knob: arms when turned on; only direct sound-knob edits start eight seconds of learning, then it idles	Off to On Default: Off
Bypass	Plays the input as it is (the POWER switch on the amplifier)	Off to On Default: Off

Presets and files

Presets are browsed with the arrows or the preset-name menu. A star marks an edited patch. Save supports folders, Rename, Delete and opening the user preset folder; dropping a compatible preset file onto the bar loads it. A newer-file indication means this version could not apply every setting.

The host session stores the complete patch. A product preset can deliberately leave listening trim, audition or display controls unchanged; the product-specific rules below describe those exceptions.

Gear Tube presets use .geartube in ~/Library/Audio/Presets/Barakigal Audio/Gear Tube. Power/Bypass and Auto Gain are excluded from presets; Output belongs to the processing sound and is included.

Installation and help

Download the macOS archive from the product page and quit your audio host. Place the AU .component in ~/Library/Audio/Plug-Ins/Components, VST3 in ~/Library/Audio/Plug-Ins/VST3, or CLAP in ~/Library/Audio/Plug-Ins/CLAP. The product page also offers an installer with format selection. Restart or rescan the host. To uninstall, quit the host and remove that product's bundles from those folders; saved sessions and user presets are separate.

For questions or a problem report, use BRK Audio support on GitHub. Include the plugin version, macOS version, host and format, sample rate and a short description of how to reproduce it.